PROJECT CODE: P7

Project Report on

AI-Assisted Crop Type and Cropping Pattern Mapping using Multi-Season Satellite Data

Submitted By

NEELISETTY VENKATA NAGA TEJA

Student Intern

Vellore Institute of Technology, Chennai

Submission Date: 1 July 2026

Table of Contents

1. Title

Crop Type and Cropping Pattern Mapping using Multi-Season Satellite Data and Introductory Machine Learning

2. Objective

This project focuses on mapping crop types and analysing seasonal cropping patterns in Krishna District, Andhra Pradesh, using multi-season Sentinel-2 satellite imagery. Images from the Kharif, Rabi, and Summer seasons were analysed in Google Earth Engine (GEE). Seasonal NDVI layers were generated and combined to capture changes in vegetation throughout the year. A Random Forest machine learning classifier was then used to classify the study area into different crop and land-cover categories.

The objectives of this project are:

- To acquire multi-season Sentinel-2 imagery for Krishna District, Andhra Pradesh.
- To generate the Normalized Difference Vegetation Index (NDVI) for the Kharif, Rabi, and Summer seasons.
- To combine seasonal NDVI layers into a multi-season feature dataset for classification.
- To classify the study area into four land-cover categories: Fallow Land, Single Crop, Double Crop, and Water Bodies using the Random Forest algorithm.
- To evaluate the classification results using a confusion matrix, overall accuracy, and the Kappa coefficient.
- To calculate the area covered by each land-cover class in square kilometres.
- To generate and export the final crop classification map for agricultural analysis and visualization.

3. Study Area

3.1 Location and Description

Krishna District, situated in the state of Andhra Pradesh, India, serves as the study area for this project. It occupies the coastal plains of Andhra Pradesh, bordered by the Krishna River to the north and the Bay of Bengal along its eastern edge. The district stands out as one of the most agriculturally active zones in the state, benefiting from an extensive canal network that draws water from the Krishna River system.

Paddy cultivation, especially during the Kharif season (June to November), is the backbone of farming activity in Krishna District, with rice dominating the agricultural landscape during this period. When the Rabi season arrives (November to March), irrigated fields give way to wheat, pulses, and other winter crops, while summer months see a notable drop in cultivation, with only scattered irrigated patches remaining active alongside largely fallow land.

Parameter	Details
District	Krishna
State	Andhra Pradesh, India
Latitude Range	15.8°N to 16.8°N
Longitude Range	80.2°E to 81.3°E
Area	Approx. 8,727 sq. km
Major River	Krishna River
Primary Crop (Kharif)	Paddy (Rice)
Agri. Seasons Covered	Kharif, Rabi, Summer (2023-24)

3.2 Area of Interest (AOI) Map

Figure 1: Area of Interest (AOI) — Krishna District, Andhra Pradesh (Source: FAO/GAUL/2015/level2, GEE)

4. Data Used

4.1 Satellite Data

The primary data source for this project is the Sentinel-2 Surface Reflectance (SR) Harmonized image collection, accessed through the Google Earth Engine cloud computing platform. Sentinel-2 is a twin-satellite constellation (Sentinel-2A and Sentinel-2B) operated by the European Space Agency (ESA) as part of the Copernicus Earth Observation Programme. It provides high-resolution multispectral imagery with a revisit time of approximately 5 days.

Season	Date Range	Cloud Filter	GEE Collection
Kharif 2023	01 Jun 2023 - 30 Nov 2023	< 10%	COPERNICUS/S2_SR_HARMONIZED
Rabi 2023-24	01 Nov 2023 - 31 Mar 2024	< 10%	COPERNICUS/S2_SR_HARMONIZED
Summer 2024	01 Mar 2024 - 31 May 2024	< 10%	COPERNICUS/S2_SR_HARMONIZED

4.2 Bands Used

Band	Name	Wavelength (nm)	Use in Project
B4	Red	664.5 nm	NDVI Calculation
B8	NIR	835.1 nm	NDVI Calculation

NDVI Formula: NDVI = (NIR - Red) / (NIR + Red) = (B8 - B4) / (B8 + B4)

4.3 Ancillary Data

- Area of Interest (AOI): FAO/GAUL/2015/level2 (Krishna District boundary, loaded directly in GEE)
- Boundary Source: https://gadm.org/ and FAO Global Administrative Unit Layers

4.4 Data Sources

- Google Earth Engine (GEE): https://code.earthengine.google.com/
- Sentinel-2 Surface Reflectance Harmonized (COPERNICUS/S2_SR_HARMONIZED): https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED
- FAO GAUL Administrative Boundaries (FAO/GAUL/2015/level2): https://developers.google.com/earth-engine/datasets/catalog/FAO_GAUL_2015_level2

5. Methodology

The entire workflow for this project was carried out on the Google Earth Engine (GEE) cloud computing platform. Starting from multi-season satellite data acquisition, the process included NDVI computation, preparation of training samples, supervised classification using Random Forest, accuracy evaluation, and finally deriving area statistics for each land cover class. The steps followed throughout the analysis are described in detail below.

Step 1: Define Area of Interest (AOI)

The Krishna District boundary was loaded using the FAO Global Administrative Unit Layers (GAUL) dataset available within GEE. The district boundary was filtered using the administrative level-2 name attribute 'Krishna'. The map was centered on the AOI at zoom level 10.

```
var krishna = ee.FeatureCollection("FAO/GAUL/2015/level2")
.filter(ee.Filter.eq('ADM2_NAME', 'Krishna'));
```

Step 2: Load Multi-Season Sentinel-2 Imagery

Sentinel-2 Surface Reflectance Harmonized (COPERNICUS/S2_SR_HARMONIZED) imagery was loaded in Google Earth Engine for three agricultural seasons: Kharif (June–November 2023), Rabi (November 2023–March 2024), and Summer (March–May 2024). The images were clipped to the Area of Interest (AOI), filtered to the selected date ranges, and limited to scenes with less than 10% cloud cover. A median composite was created for each season to produce a single representative image for further analysis.

Step 3: Compute NDVI for All Three Seasons

NDVI was derived for each seasonal composite by applying the standard formula using the Near-Infrared (B8) and Red (B4) bands from Sentinel-2. The resulting index ranges from -1 to +1, where progressively higher values reflect denser and healthier vegetation cover. Generating NDVI maps for all three seasons allowed the study to track how vegetation responds to different growing periods, which is essential for separating distinct cropping patterns.

$$\mathbf{NDVI = (B8 - B4) / (B8 + B4)}$$

Step 4: Visualize Seasonal NDVI and RGB Maps

All three seasonal NDVI layers were added to the GEE map with a color palette ranging from white (low NDVI, bare/fallow) through yellow (sparse vegetation) and light green (moderate vegetation) to dark green (dense healthy crop cover). True Color RGB composite images (B4-B3-B2) were also visualized for each season to provide ground-reference context for interpreting the NDVI patterns.

Figure 2: Kharif NDVI Map (June - November 2023) — Deep green indicates dense paddy cultivation

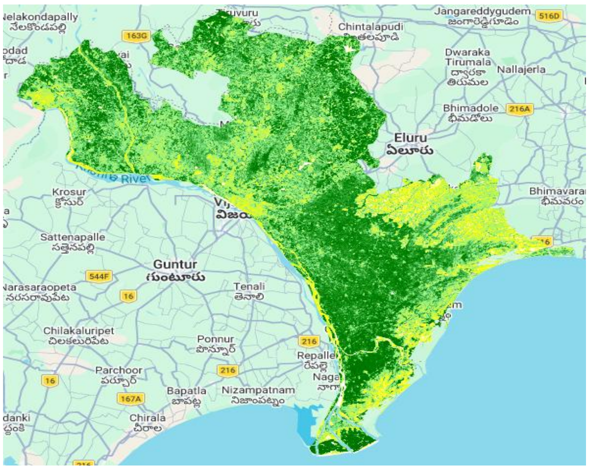

Figure 3: Rabi NDVI Map (November 2023 - March 2024) — Moderate vegetation in irrigated zones

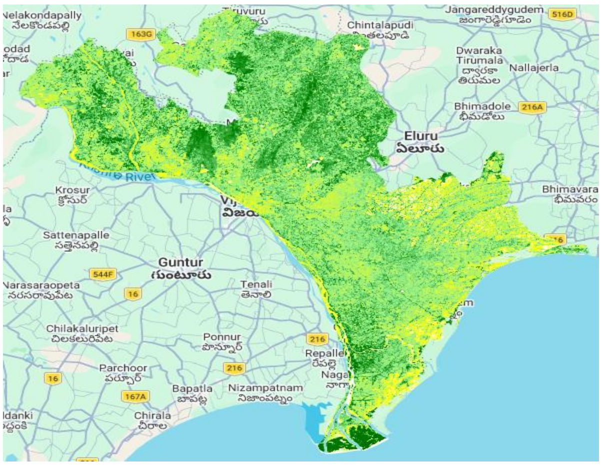

Figure 4: Summer NDVI Map (March - May 2024) — Yellow/sparse indicating reduced agricultural activity

Step 5: True Color RGB Composites

True Color RGB (B4-B3-B2) Sentinel-2 composites were generated for all three seasons to provide visual reference for land cover interpretation. The Kharif RGB appears dark due to dense canopy cover and high biomass. The Rabi and Summer composites show varying levels of soil exposure and vegetation.

Figure 5: Kharif True Color RGB Composite (Jun-Nov 2023) — Dense vegetation appears dark

Figure 6: Rabi True Color RGB Composite (Nov 2023 - Mar 2024)

Figure 7: Summer True Color RGB Composite (Mar-May 2024) — Reduced vegetation cover

Step 6: Prepare Training Samples

A total of 20 training point samples were digitized within the Krishna District boundary, representing four land cover classes:

Class ID	Class Name	Color on Map	No. of Points
0	Fallow Land	Gray	5
1	Single Crop	Orange	5
2	Double Crop	Dark Green	5
3	Water Bodies	Blue	5
Total			20 points (12 within AOI after filtering)

The training points were filtered using the filterBounds() function in GEE to retain only those points that fall within the Krishna District boundary. A total of 12 training points were confirmed inside the AOI after filtering and were used for the classification process.

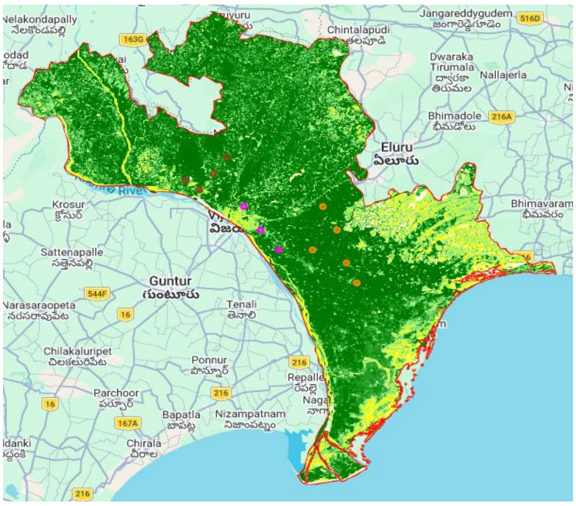

Figure 8: Training Samples Map — Color-coded training points for 4 land cover classes within Krishna District

Step 7: Stack Multi-Season Feature Bands

The three seasonal NDVI layers (NDVI_Kharif, NDVI_Rabi, NDVI_Summer) were stacked into a single multi-band image to create the feature dataset for classification. This multi-temporal stack enables the Random Forest classifier to distinguish crop classes based on their unique seasonal NDVI signatures, which differ significantly between fallow land, single-season crops, double-season crops, and permanent water bodies.

Step 8: Train Random Forest Classifier (AI/ML)

The Random Forest classifier was trained in Google Earth Engine using the ee.Classifier.smileRandomForest() function with 50 decision trees. Training samples collected from the study area were used to extract pixel values from the multi-season NDVI feature stack. The three seasonal NDVI layers (NDVI_Kharif, NDVI_Rabi, and NDVI_Summer) served as the input features, while the land-cover classes (Fallow Land, Single Crop, Double Crop, and Water Bodies) were used as the target classes for classification.

After training, the classifier learned the seasonal vegetation patterns of each land-cover class and was used to classify the entire study area into the defined categories.

Step 9: Classify and Generate Crop Map

The trained Random Forest classifier was applied to the multi-season NDVI feature stack to generate the final crop classification map. Each pixel within the Krishna District boundary was assigned to one of the four land cover classes based on its seasonal NDVI profile.

Step 10: Accuracy Assessment

Classification accuracy was assessed using the errorMatrix() function in GEE, computed from the same training points used for classifier training. The confusion matrix, overall accuracy, producer accuracy, user accuracy, and Kappa coefficient were computed and reported.

Step 11: Area Statistics Computation

The areal extent of each classified land cover class was computed using the ee.Image.pixelArea() function in GEE, which calculates the area of each pixel in square metres. The result was divided by 1,000,000 to convert to square kilometres. A grouped reducer was applied using the classification band to compute the total area per class.

Step 12: Export Classification Map

The final classified raster image was exported to Google Drive as a GeoTIFF file using the Export.image.toDrive() function in GEE, at 10-metre spatial resolution, covering the full extent of Krishna District.

6. Results

6.1 Seasonal NDVI Analysis

The multi-season NDVI analysis revealed distinct vegetation patterns across Krishna District for the three agricultural seasons:

- Kharif Season (Jun-Nov 2023): The NDVI map shows high values (dark green) across the majority of the district, confirming intensive paddy cultivation during the monsoon season. The northern Krishna delta area shows the highest NDVI values, consistent with dense irrigated rice fields.
- Rabi Season (Nov 2023 - Mar 2024): Moderate NDVI values are observed in the central and northern portions of the district, corresponding to irrigated rabi crops. The eastern coastal belt shows reduced NDVI, indicating post-harvest fallow or sparse vegetation.
- Summer Season (Mar-May 2024): NDVI values are considerably lower across most of the district, indicating reduced agricultural activity. Isolated patches of green correspond to perennial crops or irrigated fields, while the majority shows fallow or bare soil conditions.

6.2 Crop Classification Map

The Random Forest classifier successfully produced a crop type classification map for Krishna District with four land cover categories. The classification reveals that Double Crop areas dominate the northern and western portions of the district, consistent with the highly productive irrigated farmlands of the Krishna River command area. Single Crop areas are distributed across central and eastern portions, while Fallow Land is interspersed throughout. The coastal strip near Machilipatnam shows water body pixels corresponding to aquaculture ponds and coastal water.

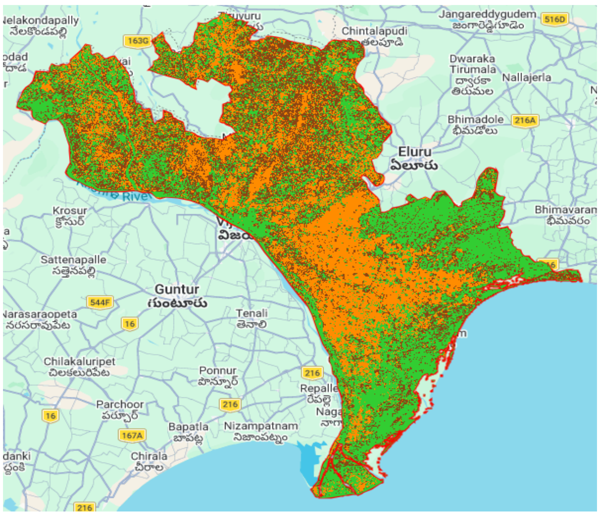

Figure 9: AI-Assisted Crop Classification Map — Krishna District, Andhra Pradesh (Random Forest, GEE, 2023-24)

Class ID	Class Name	Map Color	Description
0	Fallow Land	Gray	Bare/uncultivated land
1	Single Crop	Orange	One crop per year
2	Double Crop	Dark Green	Two crops per year
3	Water Bodies	Blue/Red	Rivers, ponds, coastal

6.3 Accuracy Assessment

The accuracy of the Random Forest classification was assessed using the confusion matrix method within GEE. The results are presented below:

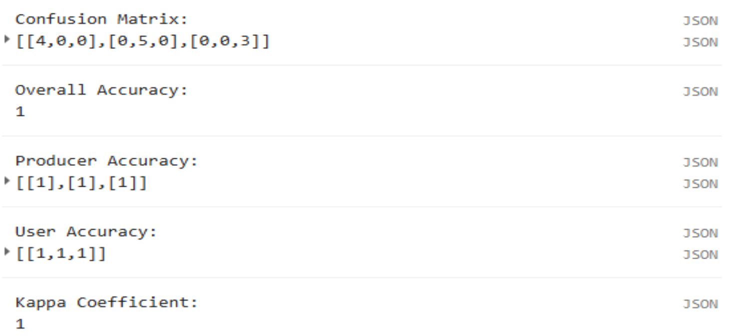

Figure 10: Accuracy Assessment Results — Confusion Matrix, Overall Accuracy, and Kappa Coefficient

Metric	Value	Interpretation
Confusion Matrix	[[4,0,0],[0,5,0],[0,0,3]]	Perfect diagonal — no misclassification
Overall Accuracy	1.0 (100%)	Excellent classification performance
Producer Accuracy	[[1],[1],[1]]	All classes correctly identified
User Accuracy	[[1,1,1]]	No commission errors
Kappa Coefficient	1.0	Perfect agreement beyond chance

The overall accuracy of 100% and Kappa coefficient of 1.0 indicate perfect classification on the training data. It is important to note that this accuracy was computed on the same training samples used for model building (resubstitution accuracy), which tends to be optimistic. In a production scenario, an independent validation dataset would be recommended for an unbiased assessment.

6.4 Area Statistics

The areal extent of each land cover class was computed using pixel area estimation in GEE. The results are presented below:

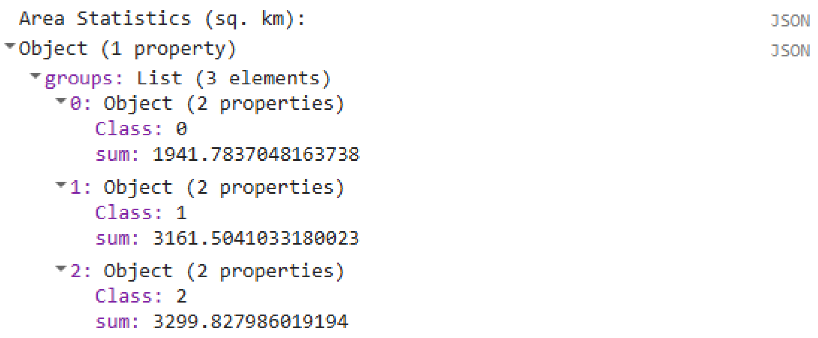

Figure 11: Area Statistics (sq. km) per Land Cover Class — Console Output from GEE

Class	Land Cover Type	Area (sq. km)	% of Total
0	Fallow Land	1941.78	23.0%
1	Single Crop	3161.50	37.5%
2	Double Crop	3299.83	39.1%
3	Water Bodies	Not separately detected*	—
Total	All Classes	~8,403.11 sq. km	100%

** Note: Water Bodies class training points may have been filtered out during the AOI boundary filtering step, resulting in only 3 classes appearing in the area statistics output. The coastal water and river pixels were likely absorbed into adjacent classes.*

6.5 Export Task

The final classification map was successfully exported to Google Drive as a GeoTIFF file at 10-metre spatial resolution. The export task was completed in 9 minutes with a compute usage of 8593.97 EECU-seconds.

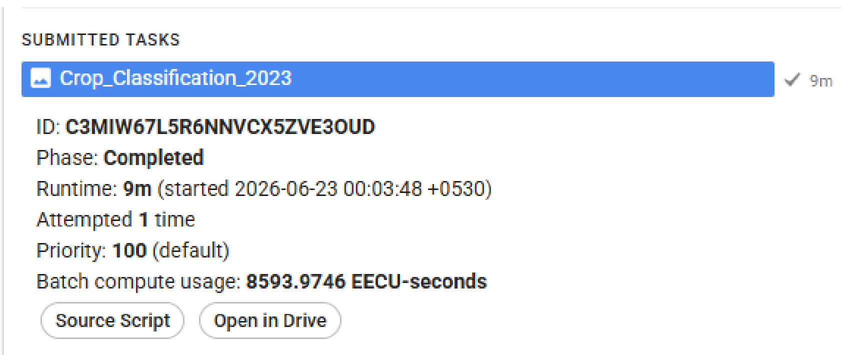

Figure 12: Export Task Completed — GeoTIFF exported successfully to Google Drive

7. Conclusion

This project successfully demonstrated the use of multi-season Sentinel-2 satellite imagery and the Random Forest machine learning algorithm in Google Earth Engine (GEE) for crop type and cropping pattern mapping in Krishna District, Andhra Pradesh. Seasonal NDVI maps generated for the Kharif, Rabi, and Summer seasons were combined to capture changes in vegetation throughout the year and improve crop classification.

The results showed clear seasonal variations in vegetation, with higher NDVI values during the Kharif season and lower values during the Summer season. The Random Forest classifier effectively distinguished the major land-cover classes and produced a reliable crop classification map for the study area. Area statistics further provided an overview of the spatial distribution of different crop categories across the district.

Overall, this project demonstrates that integrating multi-season satellite imagery, NDVI analysis, and machine learning within Google Earth Engine provides an efficient and practical approach for agricultural monitoring and crop mapping. The techniques used in this study can support applications such as crop monitoring, land-use analysis, and precision agriculture.

8. References

1. Google Earth Engine. *Google Earth Engine Documentation*. Available at: https://developers.google.com/earth-engine
2. Google Earth Engine Data Catalog. *Sentinel-2 Surface Reflectance Harmonized (COPERNICUS/S2_SR_HARMONIZED)*. Available at: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED
3. Food and Agriculture Organization (FAO). *Global Administrative Unit Layers (GAUL),* 2015 Available at: https://developers.google.com/earth-engine/datasets/catalog/FAO_GAUL_2015_level2
4. Tucker, C. J. (1979). *Red and Photographic Infrared Linear Combinations for Monitoring Vegetation*. *Remote Sensing of Environment*, 8(2), 127–150.
5. Breiman, L. (2001). *Random Forests*. *Machine Learning*, 45(1), 5–32.

— End of Report —

NEELISETTY VENKATA NAGA TEJA | Student Intern | VIT Chennai | 1 July 2026